NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Renal History and Physical Exam

Adrienne Loftis, DO, MS
Clinical Medicine Department | Arkansas Campus
aloftis@nyit.edu

Do.
Make.
Heal.
Innovate.
Reinvent the Future.

SESSION OBJECTIVES

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- **Recognize** common renal and urinary tract complaints
- **Conduct** a focused renal and urinary history and physical examination.
- **Analyze** key history and physical examination findings to **construct** prioritized differential diagnoses based on patient presentation
- **Evaluate** clinical evidence to support diagnostic decision-making.
- **Compose** clear, accurate, and organized clinical documentation.
- **Apply** clinical reasoning to solve patient case scenarios and defend diagnostic conclusions.

Source: Course Syllabus

ABBREVIATIONS

NEW YORK INSTITUTE
CHNOLOGY
of Osteopathic
ne

Common Medical Abbreviations

Abbreviation	Definition
ACEi	Angiotensin-Converting Enzyme Inhibitor
AKI/AKD	Acute Kidney Injury / Acute Kidney Disease
ARB	Angiotensin Receptor Blocker
BID	Twice Daily (Bis in Die)
CKD	Chronic Kidney Disease
CVA	Costovertebral Angle
HEENT	Head, Eyes, Ears, Nose, and Throat
LE	Lower Extremity
NAD	No Acute Distress
NC/AT	Normocephalic/Atraumatic
NSAID	Nonsteroidal Anti-Inflammatory Drug
OSE	Osteopathic Structural Exam
PND	Paroxysmal Nocturnal Dyspnea
RBC	Red Blood Cell
TMP-SMX	Trimethoprim-Sulfamethoxazole
UA	Urinalysis
UVJ	Ureterovesical Junction
WBC	White Blood Cell

Generated with the use of Microsoft CoPilot

CLINICAL CASE #1

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine



Image generated using Microsoft CoPilot

- *24-year-old woman present with 2 days of gradually, worsening, constant painful urination.*
  - *Achy at the stream*
  - *2/10 without radiation*
- *Lower ABD pressure*
- *New sexual partner, no barrier contraception*
- *Admits urgency*
- *No fever, chills, back pain, nausea, vomiting, or hematuria.*

POLL QUESTION

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine



RENAL SYSTEM

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Upper Urinary Tract

- Kidneys (2)

  Filter blood, regulate fluid/electrolytes, acid-base balance, blood pressure regulation, produce hormones

- Ureters (2)

  Transport urine from kidneys to bladder

Lower Urinary Tract

- Bladder

  Stores urine until voiding

- Urethra

  Transport urine to outside of the body

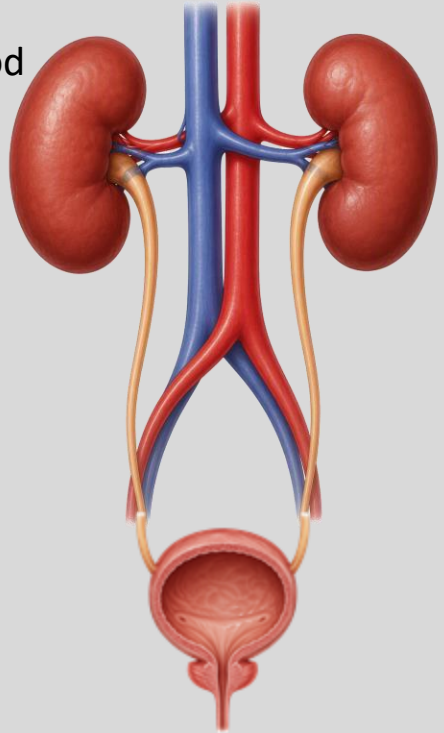


Image generated using Microsoft CoPilot

FOCUSED RENAL HISTORY- ASSOCIATED SYMPTOMS

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

URINE

- Dysuria
- Urinary frequency
- Urinary urgency
- Hematuria
- Nocturia
- Change in stream
- Incontinence

PAIN



- Flank pain
- CVA pain
- Suprapubic pain
- Neurologic pain

- Radiation of pain
- Onset of pain
- Duration of pain
- Severity of pain

SYSTEMIC



- Fever / chill
- Nausea / vomiting
- Fatigue
- Decreased appetite
- Weight changes
- Peripheral edema

FUNCTIONAL



- Polyuria
- Oliguria
- Polydipsia
- Nocturia
- Weak stream
- Intermittency

Use open-ended questions first, then narrow as needed.

FOCUSED RENAL HISTORY- REVIEW OF SYSTEMS

NEW YORK INSTITUTE

Focus on systems that help identify causes, complications, and associated conditions that affect renal function.

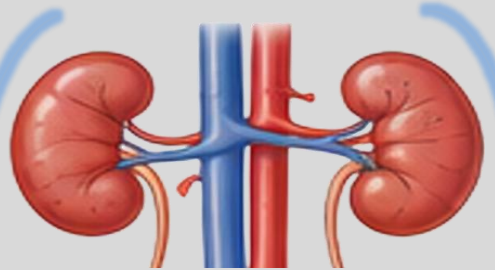
CONSTITUTIONAL
- Fever, chills, night sweats
- Fatigue
- Headache
- Unintended weight changes

RESPIRATORY
- Shortness of breath/dyspnea
- Cough
- Pulmonary edema

GASTROINTENSTINAL
- Nausea/vomiting
- Abdominal or flank pain
- Constipation or diarrhea
- Loss of appetite
- Metallic taste

NEUROLOGIC
- Altered mental status
- Confusion
- Dizziness, lightheadedness
- Difficulty concentrating
- Seizures

GENITOURINARY
- Changes in urination: frequency, urgency, hesitancy, straining
- Dysuria, hematuria
- Nocturia, enuresis
- Polyuria, oliguria, anuria
- Incontinence
- Dribbling



RENAL-FOCUSED REVIEW OF SYSTEMS

CARDIOVASCULAR
- Peripheral edema
- Shortness of breath
- Orthopnea
- Paroxysmal nocturnal dyspnea
- Palpitations

INTEGUMENTARY
- Pruritis
- Purpura
- Dry skin, rash
- Pallor
- Livedo reticularis

HEMATOLOGIC
- Easy bruising or bleeding
- Abnormal clotting

MUSCULOSKELETAL
- Muscle cramping
- Joint pain or swelling

ENDOCRINE
- Heat/cold intolerance
- Polydipsia

SIGNIFICANT MEDICAL HISTORY

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Medical: Cardiac disorder/disease/failure, immunocompromised, endocrine dysfunction, neurological abnormality, renal insufficiency, liver disease, sexual activity, contraception, pregnancy
- Surgical: abdominal, genitourinary, renal, urinary
- Social: habits, diet
- Medications: NSAIDs, diuretics, ACEi/ARB, radiocontrast
- Allergies: sulfa, reactions

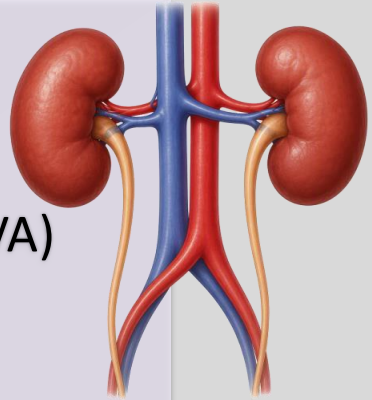
LOCALIZE THE PROBLEM

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Localization can help narrow your differential and guide your examination

UPPER TRACT

- Fever/chills
- Flank pain
- Costovertebral angle (CVA) tenderness
- Nausea/vomiting
- Hematuria



LOWER TRACT

- Dysuria
- Urinary frequency
- Urinary urgency
- Suprapubic tenderness



Images generated using Microsoft CoPilot

DIFFERENTIAL DIAGNOSIS FRAMEWORK

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

INFECTIOUS	OBSTRUCTIVE	GLOMERULAR / PARENCHYMAL	OTHER
• Cystitis	• Nephrolithiasis	• Glomerulonephritis	• Interstitial cystitis
• Pyelonephritis	• Benign prostatic hyperplasia	• Nephritic syndrome	• Malignancy
• Prostatitis	• Strictures	• Acute kidney infection	• Urinary retention
• Asymptomatic bacteruria	• Congenital (Horseshoe kidney)	• Chronic kidney disease	• Neurogenic bladder

Kasper DL, Fauci AS, Hauser SL, Longo DL, Jameson J, Loscalzo J. eds. Chapter 144: Urinary Tract Infections and Interstitial Cystitis. In: *Harrison's Manual of Medicine*. 19th Edition. New York, NY: McGraw-Hill; 2016.

PHYSICAL EXAMINATION

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

1. Vitals
2. General appearance
3. HEENT
4. Skin
5. Musculoskeletal
   - CVA exam
   - Extremities
6. Abdominal exam including flank
7. Genitourinary exam, as indicated

PEARL

Always explain what you are doing and why.

Maintain privacy and drape appropriately.

PHYSICAL EXAMINATION- KIDNEY

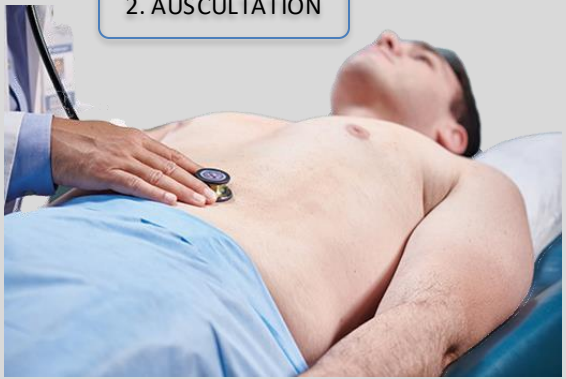
NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

1. INSPECTION



2. AUSCULTATION





3. PALPATION



4. PERCUSSION

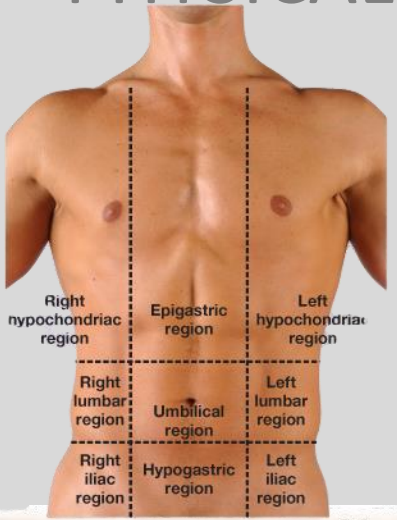


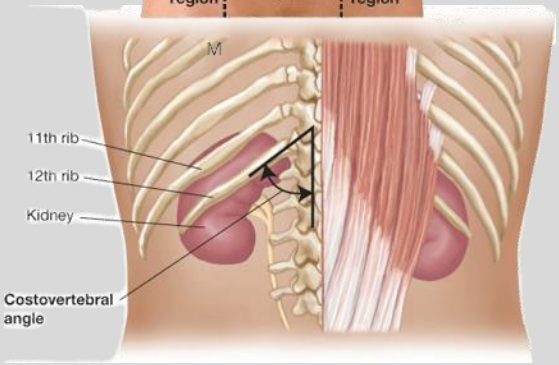
5. CVA EXAM

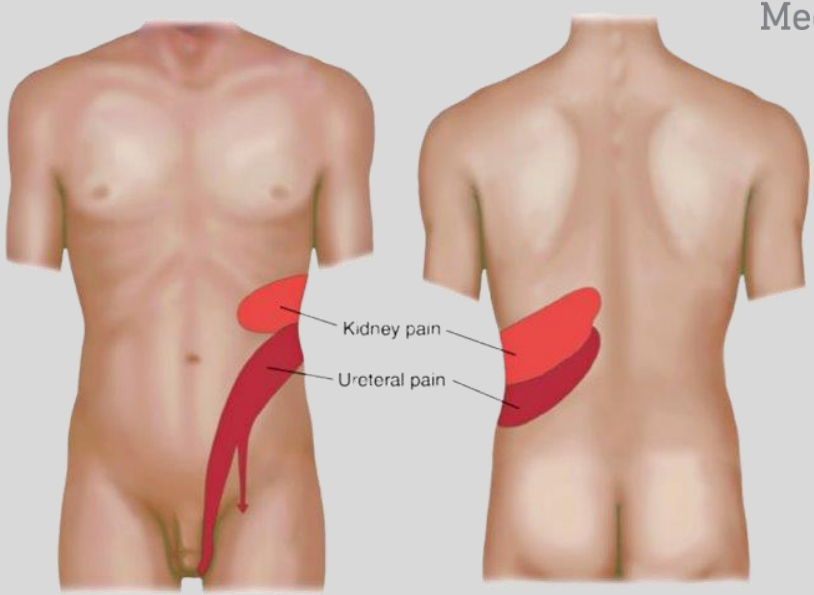
PHYSICAL EXAMINATION- KIDNEY

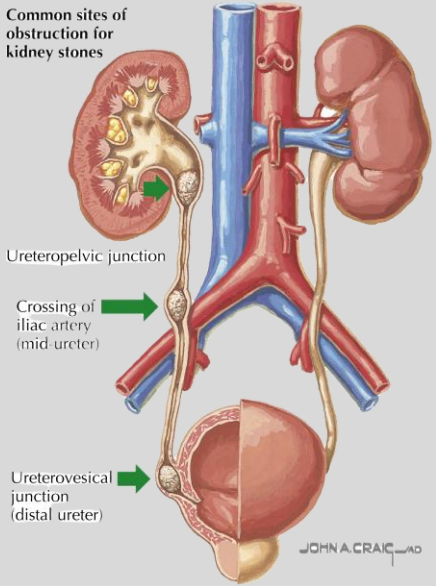
NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine









Bickley L, Chapter 19: The Abdomen. In: Bickley L, *Bates' Pocket Guide to Physical Examination and History Taking*. 9e. Philadelphia, PA: Wolters Kluwer; 2021.

PHYSICAL EXAMINATION- BLADDER

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

INSPECTION

Distended bladder may be visible or cause lower abdominal bulging.

PALPATION

Palpate suprapubic area for tenderness or bladder distention.

PERCUSSION

Dullness over suprapubic area suggests bladder distention

PEARL

Always ask permission before palpation.

Be gentle and systematic.

Maintain privacy and drape appropriately.

EXAM FINDINGS

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

FINDINGS	SUGGESTS
Fever, chills	Pyelonephritis, systemic infection
CVA tenderness or renal colic	Pyelonephritis, nephrolithiasis
Suprapubic tenderness	Cystitis, urinary retention
Edema (periorbital or LE)	Glomerular disease, nephrotic syndrome, AKI
Hypertension	CKD, renovascular disease
Abnormalities of external urethra opening or genitalia	Infection, irritation, trauma

LABS- URINALYSIS

NEW YORK INSTITUTE OF TECHNOLOGY

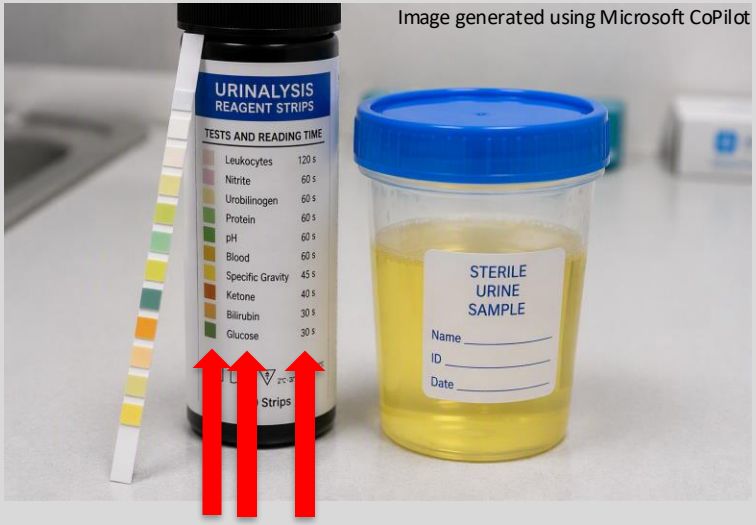


Image generated using Microsoft CoPilot

Urinalysis Finding	Clinical Significance
Color & Clarity	Hydration status and urine concentration
Leukocyte Esterase	White blood cells; suggests UTI or urinary tract inflammation
Nitrites	Gram-negative bacterial infection (commonly UTI)
Urobilinogen	Hemolysis or hepatocellular disease
Protein	Glomerular disease, CKD, nephrotic syndromes
Blood (heme/RBCs)	Stones, cysts, tumors, glomerular disease, trauma, or infection
Specific Gravity	Kidney concentrating ability and hydration status
Ketone	DKA, starvation, prolonged fasting, pregnancy
Bilirubin	Hepatobiliary disease or biliary obstruction
Glucose	Hyperglycemia or proximal tubular dysfunction

Image generated using Microsoft CoPilot

Always correlate with history, physical exam, and other labs, when indicated

RENAL LABS- SERUM AND URINE SPECIMENS

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic

Kidney Function

- Creatinine
- BUN
- eGFR

CLINICAL PEARL

Start with creatinine, eGFR, BUN, electrolytes, and urinalysis before pursuing targeted serologic testing.

Electrolytes & Acid-Base

- Potassium
- Bicarbonate
- Calcium
- Phosphorus

Microscopic Urine

- WBCs
- RBCs
- Neutrophils
- Eosinophils
- Epithelial cells
- Casts (i.e. WBC,RBC, hyaline, granular)
- Crystals

Etiology-Specific Tests

- ANA
- ANCA
- Anti-GBM
- C3/C4
- SPEP

Other Lab Tests

- Hemoglobin
- Iron Studies
- Albumin
- PTH
- Osmolality (serum or urine)
- Magnesium

RENAL IMAGING

- Ultrasound
  - With duplex Doppler
- CT
  - With contrast
  - Without contrast
  - W/w/o
- Nuclear scan

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

RETURN TO OUR PATIENT

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine



POSITIVE FINDINGS

- ✓ Dysuria
- ✓ Suprapubic discomfort
- ✓ Urinary urgency
- ✓ New sexual partner

NEGATIVE FINDINGS

- ✗ No fever or chills
- ✗ No flank pain
- ✗ No CVA tenderness
- ✗ No hematuria

MOST LIKELY DIAGNOSIS:
ACUTE UNCOMPLICATED CYSTITIS

Clinical Case #1 Answer

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Question: *Based on this chief complaint, is this more likely an upper tract or lower tract problem?*

Correction Answer: Lower Tract

Discussion: This symptom of dysuria, or painful urination, is most commonly associated with a urinary lower tract infection (acute cystitis) involving the bladder and urethra.

Other factors to consider:

- No fever
- New sexual partner
- No Chills
- Short duration of symptoms
- No flank or back pain

Clinical Case #2

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- *Now, the patient presents with 3 days of continued dysuria with urinary frequency and urgency. Over the last 24 hours has had:*
  - *Fever and chills*
  - *Nausea*
  - *Left-sided flank pain*
- *In ER:*
  - *Temperature 39.1oC(102.4oF)*
  - *Left CVA tenderness*
  - *Positive leukocyte esterase, nitrites, WBCs*

Clinical Case #2 Answer

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Question: *The patient's new presentation most likely indicates a disease of which part of the urinary tract?*

Answer: Upper Tract

Discussion: This patient initially had symptoms consistent with a lower urinary tract infection (dysuria, frequency, urgency). The subsequent development of **systemic symptoms and signs** indicates involvement of the renal pelvis and kidney parenchyma, which is characteristic of **acute pyelonephritis.**

Other factors to consider:

- Fever
- Chills
- Flank pain
- Costovertebral angle (CVA) tenderness
- Nausea/ vomiting
- General malaise

DOCUMENTATION

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

S	24 y/o female with 2 days of constant, worsening, dysuria, urinary urgency, and suprapubic pressure. Describes as 2/10, non-radiating, achy pain. Has a new sexual partner. Denies fever, chills, flank pain, hematuria, nausea, and vomiting.
O	T 98.6F BP 118/72 HR 82bpm R14rpm SazO2 99%RA BMI 24 Input/Output: not measured General Appearance: pleasant, though anxious-appearing, woman in NAD HEENT: NC/AT, sclera non-icteric, no conjunctival pallor, no nasal inflammation or bleeding SKIN: warm, dry, intact, no rashes, bruising, or erythema CV: regular, rate, and rhythm LUNGS: CTA bilaterally ABDOMEN: soft, non-distended, normoactive bowel sounds, mild suprapubic tenderness to palpation, other quadrants non-tender, no masses, kidneys not palpated, dome of bladder palpated at PS, smooth contour EXTREMITIES: no cyanosis or peripheral edema bilateral lower extremities MUSCULOSKELETAL: no CVA or back tenderness to palpation OSE:
A	Differential Diagnosis 1. Acute uncomplicated cystitis (UTI) 2. Complicated cystitis 3. Interstitial cystitis
P	UA with urine culture. No imaging needed at this time. Increase fluid intake. TMP-SMX DS 1 tablet PO BID x 3 days Return in 2-4 weeks or sooner if fever, flank pain, hematuria. nausea/vomiting, unable to tolerate medication, or worsening symptoms.

Summary Slide

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

HIGH-YIELD TAKE HOME POINTS

- ✓Start with the chief complaint and localize the problem.
- ✓Use a focused history to identify key urinary/renal symptoms.
- ✓Perform a systematic renal examination.
- ✓Interpret findings to build a prioritized differential diagnosis.
- ✓Use labs and imaging to support, not replace, clinical reasoning.
- ✓Document clearly.

CAREFUL HISTORY + TARGETED EXAMINATION = BETTER CLINICAL DECISIONS

REQUIRED RESOURCES

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Bickley L, Chapter 19: The Abdomen. In: Bickley L, *Bates' Pocket Guide to Physical Examination and History Taking*. 9e. Philadelphia, PA: Wolters Kluwer; 2021.
- Kasper DL, Fauci AS, Hauser SL, Longo DL, Jameson J, Loscalzo J. eds. Section 10: Renal Diseases. In: *Harrison's Manual of Medicine*. 19th Edition. New York, NY: McGraw-Hill; 2016.
- **Cystitis and Pyelonephritis**
- **Renal Stones**
- **Hematuria**
- **Chronic Kidney Disease**
- **Acute Kidney Injury**
- **Renal Laboratory Tests and Urinalysis**

HOW TO CONTACT

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Student Hours: Fridays 11a-12p (CT)/ 12p-1p (ET)

- In-Person: Clinical Medicine Department (424D)
- Zoom: https://nyit.zoom.us/my/aloftisdo
- Email questions: aloftis@nyit.edu

Lecture and Lab Feedback Form:

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R